Ari Hendrawan

Risk & Data Scientist · Credit Scoring · Decision Engine · Python

arih.ai.id · linkedin.com/in/arihendra2802 · github.com/Arihkun · arihendra2802@gmail.com

Risk & Data Scientist based in Jakarta specializing in credit-risk modeling, Credit Decision Engine (CDE) rule architecture, and automated portfolio analytics in Indonesia's multi-finance industry. Experienced in translating lending policies and historical credit datasets into validated scorecards, GoRules/ Zen decision models, and production software.

SELECTED PROJECTS & TECHNICAL DOCUMENTATION

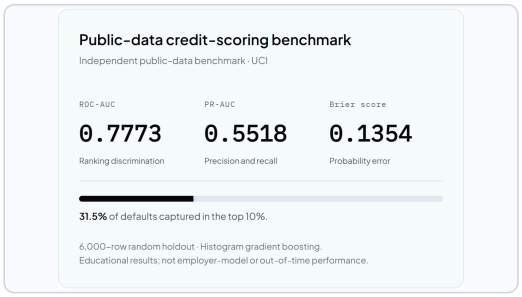
Credit Scoring ML & Risk Modeling — Risk Development · Scorecard & Holdout Benchmark

Python · XGBoost · LightGBM · scikit-learn · Optuna · ROC-AUC / KS / Gini

End-to-end credit-risk modeling workflow built to evaluate borrower default risk, enforce application-time feature availability, and segment applicants into actionable risk tiers for underwriting handoff.

- Developed and evaluated gradient-boosted and linear classifiers across 24,840 credit records and 28 candidate features (achieving holdout ROC-AUC 0.757, Gini 0.515, and KS 0.402 in internal development, plus 0.777 ROC-AUC on the open public benchmark).
- Audited feature lineage to eliminate post-application data leakage, analyzed decile bad-rate separation, and prepared FastAPI input schemas for IT integration.

↗ arih.ai.id/projects/credit-scoring-ml



Model benchmark comparison (LightGBM, XGBoost, Logistic Regression) & KS/ROC-AUC evaluation metrics.

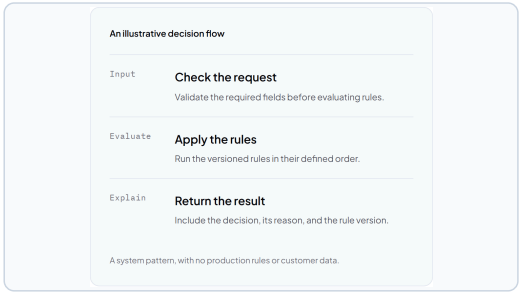
Credit Decision Engine (GoRules / Zen) — Risk Development · Automated Underwriting Rules

GoRules Zen · JSON Decision Model · Python · FastAPI · Pydantic · pytest

Modular decision-engine architecture translating multi-finance credit policy matrices into versioned GoRules/Zen decision tables with deterministic reason codes and regression testing.

- Separated payload normalization, reference lookup mapping, rule-table evaluation, and response assembly so policy changes can be updated and audited independently.
- Built boundary and parity test suites in Python/pytest to verify rule execution order, knock-out conditions, and API output contracts before production deployment.

↗ arih.ai.id/projects/cde-gorules



Multi-stage Credit Decision Engine flow: applicant normalization, GoRules policy matrix, and decision routing.

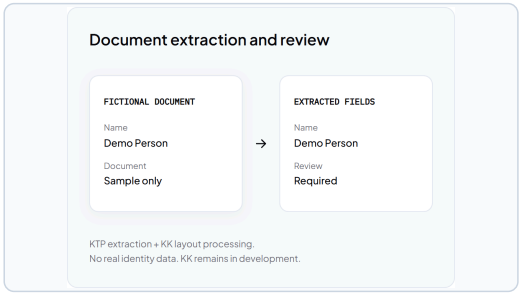
Indonesian Document OCR (KTP & Kartu Keluarga) — Applied AI Engineer · Identity Extraction Pipeline

PyTorch · LayoutLM · OpenCV · FastAPI · Confidence Routing · Human-in-the-Loop

Document intelligence service designed to extract, normalize, and validate structured identity fields from Indonesian KTP and Kartu Keluarga (KK) images during customer onboarding.

- Combined image preprocessing, OCR token detection, and field-level pattern checks (NIK 16-digit structure, date consistency, and regional prefix validation).
- Implemented per-field confidence scoring that automatically routes low-confidence or blurred fields to a human review queue rather than silently passing uncertain data.

↗ arih.ai.id/projects/indonesian-document-ocr



Structured KTP/KK field extraction preview with per-field confidence scores and validation status.

SHIPPED WEB APPLICATIONS & PRODUCT ENGINEERING

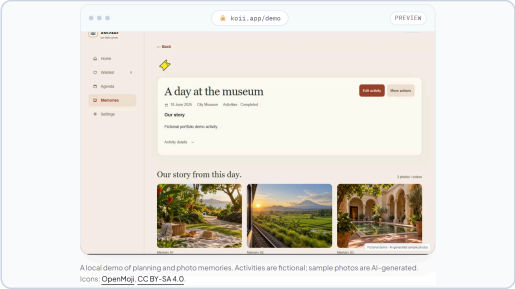
Koii — Shared Planning & Memory Space — Full-Stack Creator · Progressive Web Application

Next.js | TypeScript | Supabase | Google Calendar API | Cloudflare Workers | Web Push

Private two-person web application for managing shared wishlists, scheduling dates with two-way Google Calendar sync, and archiving photo memories with custom stickers.

- Engineered invite-code pairing, Row-Level Security (RLS) workspace isolation in Supabase, and automated Web Push reminders via Cloudflare Workers cron triggers.
- Built client-side WebP image compression and polaroid/sticker canvas rendering for fast mobile uploads and low storage overhead.

↗ koii-app.pages.dev · arih.ai.id/projects/koii



Koii interface showcasing shared activity planning, calendar sync, and memory polaroid cards.

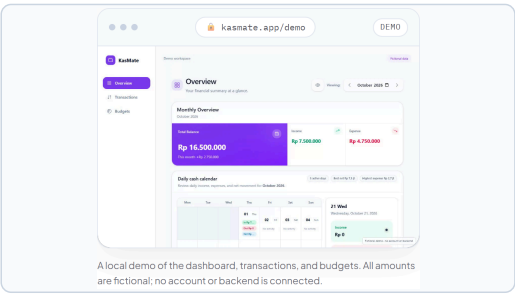
KasMate — Personal Finance & Receipt Tracker — Full-Stack Creator · Local-First Finance App

React | TypeScript | Tailwind CSS | Receipt OCR | Local-First Storage

Everyday money management application uniting multi-wallet tracking, recurring bill reminders, monthly category budgets, and receipt scanning in a privacy-first interface.

- Designed cashflow breakdown charts, budget burn-rate alerts, and multi-account balance reconciliation for daily personal expense tracking.
- Added structured receipt entry and exportable monthly summaries with privacy masking controls.

↗ kasmate.pages.dev · arih.ai.id/projects/kasmate



KasMate dashboard displaying wallet balances, category budget tracking, and transaction analytics.

ADDITIONAL ANALYTICS & AUTOMATION SYSTEMS

Portfolio Risk & Monthly Reporting Automation

Python | Polars | Parquet | DuckDB | Excel/VBA

Staged columnar data pipeline that separates raw source workbooks from reusable Parquet analytical feeds and DuckDB queries, feeding automated Excel/VBA monthly portfolio reports with contract-level mapping reconciliation. ↗ Read case study

Repeat-Order Propensity & Vintage Analytics

Python | SQL Server | Temporal Validation | Cohort Analysis

Agreement-level repeat-order propensity workflow and Vintage/Roll-Rate delinquency tracking system with strict out-of-time validation and leakage checks to support multi-finance retention and risk reviews.

PROFESSIONAL EXPERIENCE & EDUCATION

Risk Development Staff — PT Smart Multi Finance (Jakarta, Indonesia) — Sep 2025 – Present

- Prepare and validate credit and portfolio datasets, develop and evaluate credit-scoring models (XGBoost/LightGBM), and design GoRules/Zen decision tables for automated underwriting.
- Produce monthly Credit Scoring and CDE performance monitoring reports, build Polars/DuckDB/Parquet reporting pipelines, and investigate production data discrepancies with IT and business teams.

Bachelor of Informatics (S.Kom.) — University of Technology Yogyakarta · GPA 3.32 / 4.00 — 2020 – 2024

AREAS OF EXPERTISE

Credit Risk & Modeling

Scorecard Development, AUC/Gini/KS Evaluation, Risk Tier Segmentation, Temporal Validation, XGBoost, LightGBM, Optuna

Decision Systems & Data

GoRules Zen Engine, JSON Decision Models, Python, Polars, DuckDB, Parquet, SQL Server, Excel/VBA Automation

Applied AI & Web Delivery

Document OCR (KTP/KK), FastAPI, Next.js, React, TypeScript, Supabase, Cloudflare Pages & Workers

*Full interactive portfolio, live application demos, and technical write-ups available at **arih.ai.id***